4.3.1 Relational structures

Relational structures organize data into normalized tables with defined relationships between entities. This approach excels at maintaining data consistency, supporting complex queries across multiple entities, and handling updates to discrete records.

The relational model's strength lies in query flexibility—you don't need to anticipate every possible question when designing the schema. Unlike other approaches that optimize for specific access patterns, relational databases excel at ad-hoc analytical queries that weren't considered during initial design. This flexibility becomes invaluable as digital twin requirements evolve and new analytical needs emerge.

Figure 4.6 shows a simplified example of how maintenance tasks for an industrial asset can be represented as a relational data model, with each table holding rows representing instances of an entity and key references between the tables representing connections between these entities. In this way, however many workorders and tasks are created for an asset, there will only ever be one asset record, with all other entities related to it.

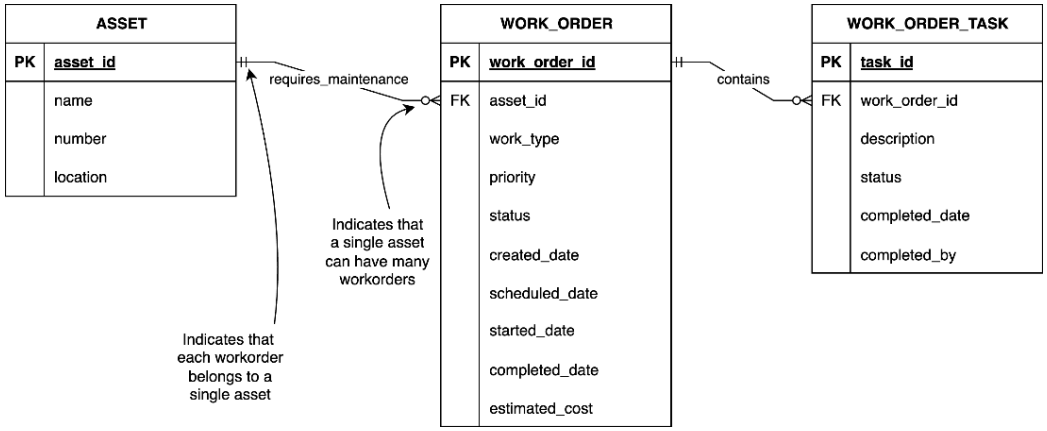


Figure 4.6 A relational model of a work order showing tables and properties representing entities, and their relationships.

This type of data store allows ad-hoc analytical queries to be executed, such as "show me all the maintenance tasks performed on-time by John Smith on the printing press with identifier F1" as shown in listing 4.3. The database for this example is provided alongside the code in GitHub.